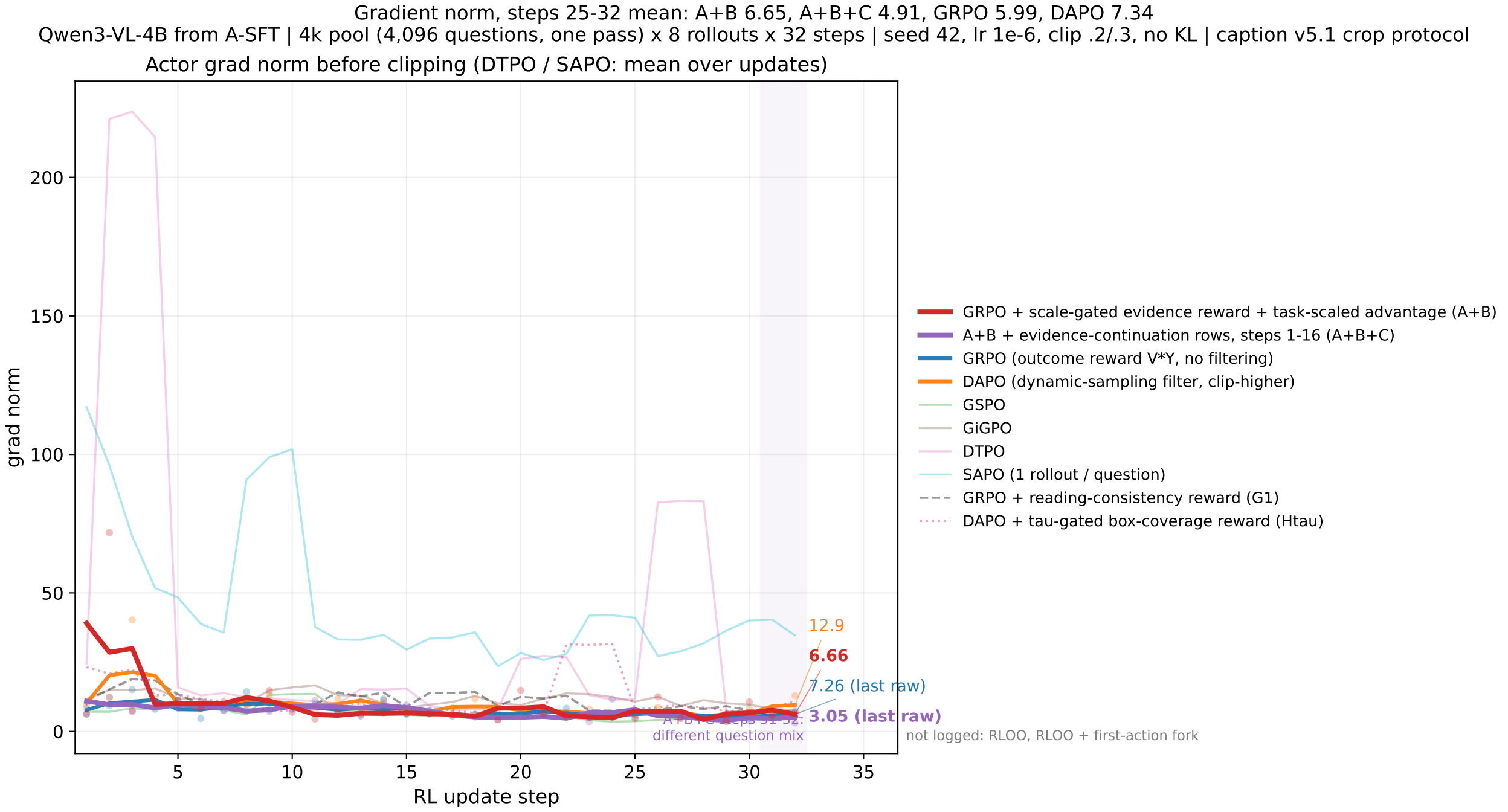
Gradient norm, steps 25-32 mean: A+B 6.65, A+B+C 4.91, GRPO 5.99, DAPO 7.34
Qwen3-VL-4B from A-SFT | 4k pool (4,096 questions, one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, clip .2/.3, no KL | caption v5.1 crop protocol
Actor grad norm before clipping (DTPO / SAPO: mean over updates)
grad norm
RL update step
0
50
100
150
200
5
10
15
20
25
30
35
12.9
6.66
7.26 (last raw)
3.05 (last raw)
different question mix
GRPO + scale-gated evidence reward + task-scaled advantage (A+B)
A+B + evidence-continuation rows, steps 1-16 (A+B+C)
GRPO (outcome reward V*Y, no filtering)
DAPO (dynamic-sampling filter, clip-higher)
GSPO
GiGPO
DTPO
SAPO (1 rollout / question)
GRPO + reading-consistency reward (G1)
DAPO + tau-gated box-coverage reward (Htau)
not logged: RLOO, RLOO + first-action fork